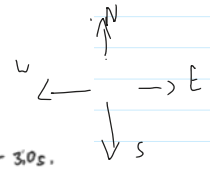
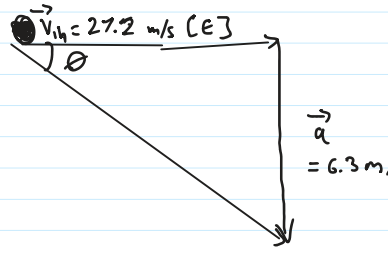
98 km/h = 27.2 m/s





12U: 2D Acceleration Problem:

1: A baseball is thrown with an initial velocity of 98 $\frac{km}{h}$ [E]. As a result of the spin on the ball, the ball accelerates ~~south~~ down at a rate of 6.3 m/s² for 3.0s.

(a) What is the final velocity of the ball

(b) What is the displacement of the ball in the 3.0s

g)

horizontal	vertical
$\vec{v}_i$ = 27.2 m/s [E]	$\vec{v}_i$ = 0
$\vec{a}$ = 0	$\vec{a}$ = 6.3 m/s²
v_f = ?	
Δt = 3 sec	Δt = 3.0 s

a) $\vec{v}_{fv} = v_i + \vec{a}\Delta t$

$v_{fv} = 0 + 6.3(3)$

$\vec{v}_{fv} =$

$= 18.9$ m/s [S]

$\phi = \tan^{-1}\left(\frac{18.9}{27.2}\right)$

$= 34.79°$

$\Delta\vec{v} = \sqrt{18.9^2 + 27.2^2}$

$= 33.698$ m/s

$\vec{v}_f = 33.7$ m/s [E 35° down]

b) Δd: $\vec{v}_{fv} = 18.9$ m/s, Δt = 3 sec

$\vec{v}_{f1} = 27.2$ m/s, Δt = 3, Δd = 81.6

$\Delta d = v_1\Delta t + \frac{1}{2}a\Delta t^2$

$\Delta d = 0(3) + \frac{6.3}{2}(3)^2$

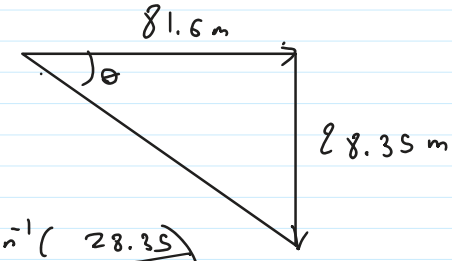
$= \frac{6.3}{2} \times 9$

$= 28.35$

$\Delta d = \sqrt{28.35^2 + 81.6^2}$

$\Delta d = 86.38$

$\Delta d_T = 86.38$ m [E 19.1° down]



$\tan^{-1}\left(\frac{28.35}{81.6}\right)$

$= 19.1$

Practice

1. A plane makes the following displacements: $\Delta\vec{d}_1$ = 72.0 km [W 30.0° S], $\Delta\vec{d}_2$ = 48.0 km [S], and $\Delta\vec{d}_3$ = 150.0 km [W]. The entire flight takes 2.5 h. T/I
   (a) Calculate the total displacement of the plane. [ans: 230 km [W 22° S]]
   (b) Calculate the average velocity of the plane. [ans: 91 km/h [W 22° S]]

$/s^2$ [d]

Notes

Vectors

↳ vectors help with references.

↳ Ex: $\vec{a} = 15\,m$ [N 75° E]

properties

↳ negative vectors:

$\vec{a} = 15\,m$ [E]

$\vec{a} = -15\,m$ [W]

↳

6 KEY EQUATIONS

1) $\Delta d = \left(\frac{v_f + v_i}{2}\right)\Delta t$
2) $v_f = v_i + a\Delta t$
3) $\Delta d = v_i\Delta t + \frac{1}{2}a\Delta t^2$
4) $v_f^2 = v_i^2 + 2a_{av}\Delta d$
5) $\Delta d = v_f\Delta t - \frac{1}{2}a\Delta t^2$

Vector Math:

↳ If vector directions are the same, you can solve algebraically

↳ If not, we need to solve with diagrams.

↳ 1) Scale diagram (Not great, difficult, inaccurate)

2) Triangle method (cosine and sine Laws, geometry Laws)

3) Component method (Trig ratios, Soh-Cah-Toa)

Practice

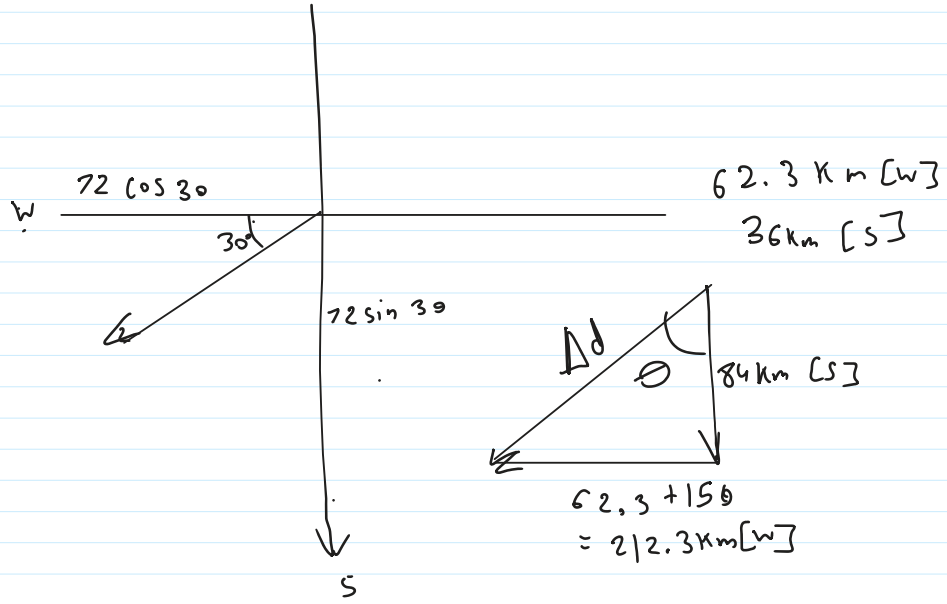
1. A plane makes the following displacements: $\Delta\vec{d}_1 = 72.0$ km [W 30.0° S], $\Delta\vec{d}_2 = 48.0$ km [S], and $\Delta\vec{d}_3 = 150.0$ km [W]. The entire flight takes 2.5 h. T/I
   (a) Calculate the total displacement of the plane. [ans: 230 km [W 22° S]]
   (b) Calculate the average velocity of the plane. [ans: 91 km/h [W 22° S]]
   (c) Calculate the average speed of the plane. [ans: 110 km/h]
2. An elk walks 25.0 km [E 53.13° N], then walks 20.0 km [S], and then runs 15.0 km [W]. The journey takes 12 h. T/I
   (a) Calculate the elk's average velocity. [ans: 0.0 km/h]
   (b) Calculate the elk's average speed. [ans: 5.0 km/h]
   (c) Explain the difference in the two answers for the elk's average velocity and average speed.

1a) $\Delta\vec{d}_1 = 72$ km [W 30 S]

$\Delta\vec{d}_2 = 48.0$ km [S]

$\Delta\vec{d}_3 = 150.0$ km [W]

$\Delta t = 2.5$



$\Delta d = \sqrt{212.3^2 + 84^2}$

$= 228.314$ km

$\theta = \tan^{-1}\left(\frac{212.3}{84}\right)$

$= 68.41$

b) $\Delta\vec{d}_T = 228.3$ km [S 68.4° W]

$\Delta t = 2.5$

$\Delta\vec{V}_{a.v} = \frac{228.3}{2.5} = 91.3$ km/h [S 68.4 W]

c) $72 + 48 + 150 = \frac{270}{2.5} =$

2)

Practice

1. A plane makes the following displacements: $\Delta\vec{d}_1 = 72.0$ km [W 30.0° S], $\Delta\vec{d}_2 = 48.0$ km [S], and $\Delta\vec{d}_3 = 150.0$ km [W]. The entire flight takes 2.5 h. T/I
   (a) Calculate the total displacement of the plane. [ans: 230 km [W 22° S]]

$\Delta d = 228.3\ km\ [S\ 68.4\ W]$

$V_{av} = 108\ km/h$

2)

Practice

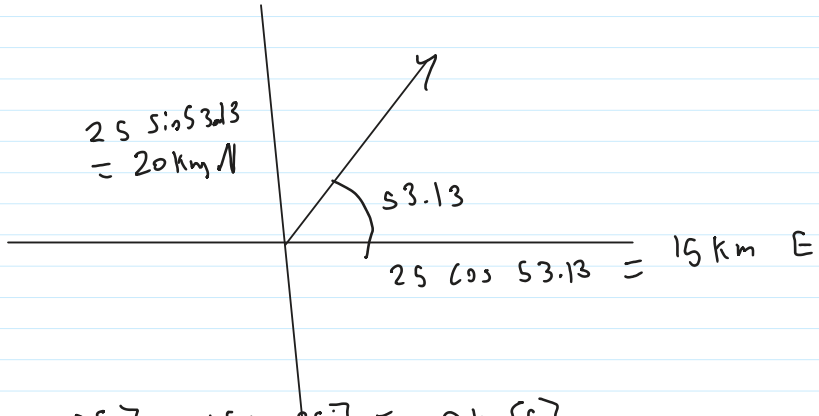
1. A plane makes the following displacements: $\Delta\vec{d}_1 = 72.0$ km [W 30.0° S], $\Delta\vec{d}_2 = 48.0$ km [S], and $\Delta\vec{d}_3 = 150.0$ km [W]. The entire flight takes 2.5 h. T/I
   (a) Calculate the total displacement of the plane. [ans: 230 km [W 22° S]]
   (b) Calculate the average velocity of the plane. [ans: 91 km/h [W 22° S]]
   (c) Calculate the average speed of the plane. [ans: 110 km/h]
2. An elk walks 25.0 km [E 53.13° N], then walks 20.0 km [S], and then runs 15.0 km [W]. The journey takes 12 h. T/I
   (a) Calculate the elk's average velocity. [ans: 0.0 km/h]
   (b) Calculate the elk's average speed. [ans: 5.0 km/h]
   (c) Explain the difference in the two answers for the elk's average velocity and average speed.

2) $\vec{d}_1 = 25$ km [E 53.13 N]

$\vec{d}_2 = 20$ km [S]

$\vec{d}_3 = 15.0$ km [W]

$\Delta t = 12$ h



15 km [E] − 15 km [E] = 0 km [E]

20 km [N] − 20 km [N] = 0 km [N]

$\vec{\Delta d} = 0$ km

∴ $\vec{V}_{av} = 0$ km

b) $V_{av} = \frac{\Delta d}{\Delta t}$ $\Delta d = 25 + 20 + 15 = 60$ km

$\Delta t = 12$ h

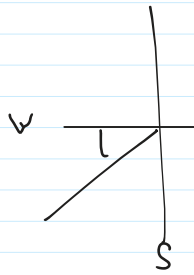
$V_{av} = \frac{60}{12} = 5$ km/h

c) Average velocity uses displacement, which is 0, as opposed

) to speed which uses distance.

Practice

1. A car heading east turns right at a corner. The car turns at a constant speed of 20.0 m/s. After 12 s, the car completes the turn, so that it is heading due south at 20.0 m/s. Calculate the car's average acceleration. T/I A [ans: 2.4 m/s² [W 45° S]]
2. Over a 15.0 min period, a truck travels at a constant speed on a road with many turns. The truck's initial velocity is 50.0 km/h [W 60.0° N]. The truck's final velocity is 80.0 km/h [E 60.0° N]. Calculate the truck's average acceleration, in kilometres per hour squared. T/I [ans: 2.80 × 10² km/h² [E 21.8° N]]



G:

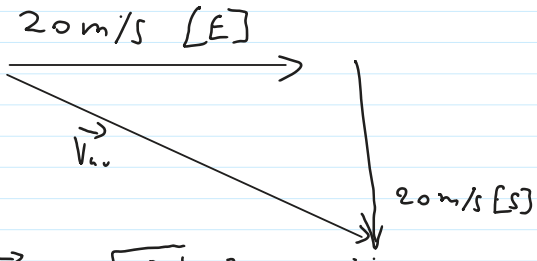
East

$v_i = 20\,m/s$

$v_{f_1} = 20\,m/s$

In turn

$v =$ constant



$\vec{v}_{av} = \sqrt{20^2 + 20^2} = 40\,m/s$